

C625

Biochemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A cook at a diner famous for breakfast fries an egg, and the clear egg white turns solid and white. The diner has been family-owned for 40 years. Which level of structure in the egg-white proteins stays unchanged?

- A. Primary structure
- B. None; all four levels, including peptide bonds, are destroyed
- C. Tertiary structure
- D. Secondary structure only; the rest is destroyed

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A research lab at a medical school with 1,000 students finds that a drug slows an enzyme, but adding much more substrate restores the normal maximum rate. The drug is taken as a pill twice a day. What type of inhibitor is the drug?

- A. Allosteric activator
- B. Irreversible inhibitor that permanently disables the enzyme
- C. Noncompetitive inhibitor
- D. Competitive inhibitor

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A noncompetitive inhibitor is added to an enzyme reaction. How do V_{max} and K_m change?

- A. Both V_{max} and K_m increase.
- B. V_{max} decreases; K_m stays the same.
- C. V_{max} stays the same; K_m increases.
- D. V_{max} always drops to zero.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A dietitian at a hospital with a new cafeteria explains that both potato starch and celery fiber are polymers of glucose. The hospital serves 2,000 meals a day. Why can humans digest the starch but not the cellulose in celery?

- A. Cellulose is a protein, not a carbohydrate.
- B. Starch has $\alpha(1\rightarrow4)$ linkages our enzymes break; cellulose has $\beta(1\rightarrow4)$ linkages we lack enzymes for.
- C. Cellulose is made of fructose instead of glucose.
- D. Cellulose has no glycosidic bonds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A nutrition teacher who runs a farmers' market booth shows olive oil, which is liquid at room temperature, next to butter, which is solid. Her booth sells local honey. What mainly explains the difference?

- A. Butter's saturated fats have more double bonds.
- B. All plant fats are liquid and all animal fats are solid.
- C. Olive oil is rich in unsaturated fatty acids whose cis double bonds keep chains from packing tightly.
- D. Olive oil has much longer fatty-acid chains.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which fatty acid has the highest melting point: 18:0, 18:1, 18:2, or 18:3?

- A. 18:0
- B. 18:3
- C. 18:2
- D. 18:1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A genetics lab at a university that sequences crop plants finds that a double-stranded DNA sample is 30% adenine. The lab's sequencer runs around the clock. What percent is guanine?

- A. 40%
- B. 30%
- C. 20%
- D. 70%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A double-stranded DNA sample is 18% cytosine. What percent is thymine?

- A. 82%
- B. 32%
- C. 64%
- D. 18%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A genetic counselor at a clinic with a long waiting list reviews a variant in which one base change turns the codon UAC (tyrosine) into UAA. The clinic recently added evening hours. What is the most likely effect on the protein?

- A. A silent mutation with no change
- B. A frameshift, since any base change shifts every later codon
- C. A missense change of one amino acid
- D. A nonsense mutation that produces a shortened protein

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A DNA template strand reads 3'-TACGGA-5'. What mRNA is transcribed from it?

- A. 5'-UCCGUA-3'
- B. 5'-UACGGA-3'
- C. 5'-AUGCCU-3'
- D. 5'-ATGCCT-3'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A biochemist at a startup with 25 employees studies a reaction with $\Delta G = +14$ kJ/mol that the cell couples to ATP hydrolysis ($\Delta G = -30.5$ kJ/mol). The startup recently raised venture funding. What is the overall ΔG , and will the coupled reaction proceed?

- A.** +14 kJ/mol; it cannot proceed
- B.** +44.5 kJ/mol; it does not proceed
- C.** -30.5 kJ/mol; it proceeds
- D.** -16.5 kJ/mol; it proceeds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

A pharmacology student preparing for a clinical rotation mixes purified phospholipids into water. Her lab partner is a former lifeguard. What structure forms?

- A.** A bilayer with heads facing the water and tails facing each other inside
- B.** A bilayer with tails facing the water and heads inside
- C.** Phospholipids dissolve evenly like table salt
- D.** Tails form hydrogen bonds with water

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A cook at a diner ~~famous for breakfast~~ fries an egg, and the clear egg white turns solid and white. ~~The diner has been family owned for 40 years.~~ Which level of structure in the egg-white proteins stays unchanged?

Target word: level unchanged by heat **Decides:** Heat does not break peptide bonds.

A	Correct	The amino-acid sequence and its peptide bonds remain.
B	Absolute language	Cooking temperatures do not hydrolyze peptide bonds.
C	Inverted operation	Tertiary folding is exactly what heat disrupts.
D	Confusion set	Secondary structure is also disrupted; only primary survives.

2. Answer D Concept 2

A research lab ~~at a medical school with 1,000 students~~ finds that a drug slows an enzyme, but adding much more substrate restores the normal maximum rate. ~~The drug is taken as a pill twice a day.~~ What type of inhibitor is the drug?

Target word: inhibitor type **Decides:** Excess substrate overcomes it: Vmax restored.

A	Inverted operation	The drug slows the enzyme; it does not activate it.
B	Adjacent category	Permanently disabled enzyme would not recover with more substrate.
C	Confusion set	Binds elsewhere and lowers Vmax; extra substrate cannot fix that.
D	Correct	It competes for the active site; enough substrate wins out.

3. Answer B Concept 2 · transfer

A noncompetitive inhibitor is added to an enzyme reaction. How do Vmax and Km change?

Target word: kinetic effect of noncompetitive inhibition **Decides:** Binds outside the active site.

A	Inverted operation	Inhibition does not raise Vmax.
B	Correct	Fewer working enzymes, same substrate affinity.
C	Confusion set	That is the competitive pattern.
D	Absolute language	Inhibition reduces rather than eliminates activity at typical doses.

4. Answer B Concept 3

A dietitian ~~at a hospital with a new cafeteria~~ explains that both potato starch and celery fiber are polymers of glucose. ~~The hospital serves 2,000 meals a day.~~ Why can humans digest the starch but not the cellulose in celery?

Target word: reason for digestibility **Decides:** α vs. β glycosidic linkage.

A	Adjacent category	Cellulose is a polysaccharide.
B	Correct	The linkage type, not the sugar, decides digestion.
C	Confusion set	The stem states both are glucose polymers.
D	Dropped element	It is held together by glycosidic bonds of a different type.

5. Answer C Concept 4

A nutrition teacher ~~who runs a farmers' market booth~~ shows olive oil, which is liquid at room temperature, next to butter, which is solid. ~~Her booth sells local honey.~~ What mainly explains the difference?

Target word: reason for melting difference **Decides:** Cis double bonds kink chains.

A	Inverted operation	Saturated means no carbon-carbon double bonds.
B	Absolute language	Coconut oil is a solid plant fat, and fish oils are liquid.
C	Correct	Loose packing lowers the melting point.
D	Adjacent category	Longer chains raise the melting point, not lower it.

6. Answer A Concept 4 • transfer

Which fatty acid has the highest melting point: 18:0, 18:1, 18:2, or 18:3?

Target word: highest melting point **Decides:** Fewest double bonds packs best.

A	Correct	A saturated chain packs most tightly.
B	Inverted operation	The most double bonds give the lowest melting point.
C	Confusion set	Two double bonds lower it further.
D	Dropped element	One double bond still adds a kink and lowers the melting point.

7. Answer C Concept 5

A genetics lab ~~at a university that sequences crop plants~~ finds that a double-stranded DNA sample is 30% adenine. ~~The lab's sequencer runs around the clock.~~ What percent is guanine?

Target word: percent guanine **Decides:** $A = T = 30\%$, so $G + C = 40\%$.

A	Dropped element	Stops at $G + C$ without splitting it.
B	Confusion set	Sets G equal to A instead of to C .
C	Correct	$100 - 60 = 40\%$ for $G + C$; $G = 20\%$.
D	Inverted operation	Subtracts only adenine from 100%.

8. Answer B Concept 5 • transfer

A double-stranded DNA sample is 18% cytosine. What percent is thymine?

Target word: percent thymine from cytosine **Decides:** $C = G = 18\%$; $A + T = 64\%$.

A	Inverted operation	Subtracts only cytosine from 100%.
B	Correct	$100 - 36 = 64$; $T = 32\%$.
C	Dropped element	$A + T$ together, not T alone.
D	Confusion set	Pairs T with C instead of with A .

9. Answer D Concept 6

A genetic counselor at a clinic ~~with a long waiting list~~ reviews a variant in which one base change turns the codon UAC (tyrosine) into UAA. ~~The clinic recently added evening hours.~~ What is the most likely effect on the protein?

Target word: mutation type and effect **Decides:** UAA is a stop codon.

A	Confusion set	Silent changes code for the same amino acid; UAA codes for none.
B	Absolute language	A substitution does not shift the reading frame.
C	Adjacent category	A stop codon ends translation rather than swapping an amino acid.
D	Correct	Translation stops early at UAA.

10. Answer C Concept 6 · transfer

A DNA template strand reads 3'-TACGGA-5'. What mRNA is transcribed from it?

Target word: mRNA from template **Decides:** Complementary, antiparallel, U for T.

A	Inverted operation	Reverses the order of the correct sequence.
B	Dropped element	Copies the template instead of pairing with it.
C	Correct	Pair T→A, A→U, C→G, G→C.
D	Confusion set	Uses T, which DNA has but RNA does not.

11. Answer D Concept 7

A biochemist at a startup ~~with 25 employees~~ studies a reaction with $\Delta G = +14$ kJ/mol that the cell couples to ATP hydrolysis ($\Delta G = -30.5$ kJ/mol). ~~The startup recently raised venture funding.~~ What is the overall ΔG , and will the coupled reaction proceed?

Target word: net ΔG of coupled reaction **Decides:** Sum the ΔG values.

A	Dropped element	Ignores the coupling to ATP.
B	Inverted operation	Subtracts the ATP value instead of adding it.
C	Wrong stem number	Reports ATP's ΔG alone.
D	Correct	$14 + (-30.5) = -16.5$; negative is spontaneous.

12. Answer A Concept 8

A pharmacology student ~~preparing for a clinical rotation~~ mixes purified phospholipids into water. ~~Her lab partner is a former lifeguard.~~ What structure forms?

Target word: structure in water **Decides:** Amphipathic molecules hide tails from water.

A	Correct	Hydrophobic tails avoid water; hydrophilic heads contact it.
B	Inverted operation	Reverses which part is water-loving.
C	Adjacent category	Their hydrophobic tails prevent full dissolving.
D	Confusion set	Hydrocarbon tails are nonpolar and do not hydrogen-bond.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Levels of protein structure	Primary structure (amino-acid sequence, peptide bonds) determines folding; denaturation by heat or pH disrupts secondary, tertiary, and quaternary structure but leaves peptide bonds intact.	Thinking denaturation breaks the amino-acid chain.	Heat, acid, or "which level is unchanged?"
2	Enzyme inhibition	A competitive inhibitor binds the active site (raises apparent K_m , V_{max} unchanged, overcome by more substrate); a noncompetitive inhibitor binds elsewhere (lowers V_{max} , K_m unchanged).	Mixing up which kinetic value each inhibitor type changes.	"Adding more substrate restores the rate."
3	Glycosidic linkages	Humans digest $\alpha(1\rightarrow4)$ linkages in starch and glycogen but lack enzymes for $\beta(1\rightarrow4)$ linkages in cellulose, though both are glucose polymers.	Thinking cellulose is made of a different sugar or is not a carbohydrate.	Starch vs. cellulose or dietary fiber.
4	Saturated vs. unsaturated fatty acids	Cis double bonds kink unsaturated chains so they pack loosely and melt lower; saturated chains pack tightly and melt higher.	Believing saturated fats contain double bonds or that plant fats are always liquid.	Oils vs. solid fats, or notation like 18:0 vs. 18:2.
5	Complementary base pairing	In double-stranded DNA, $A = T$ and $G = C$, so $A + T + G + C = 100\%$ and knowing one base gives all four.	Setting A equal to G, or subtracting one base from 100%.	A percent for one base in double-stranded DNA.
6	Transcription and mutations	mRNA is built complementary and antiparallel to the template strand with U in place of T; a point change to a stop codon (nonsense) truncates the protein.	Copying the template instead of pairing, or treating every mutation as a frameshift.	A codon change or a template sequence.
7	ATP energy coupling	An endergonic reaction can proceed when coupled to ATP hydrolysis (about -30.5 kJ/mol) if the summed ΔG is negative.	Judging a coupled reaction by the unfavorable step alone.	A positive ΔG and "coupled with ATP".
8	Phospholipid bilayer	Phospholipids are amphipathic: in water they form a bilayer with hydrophilic heads facing water and hydrophobic tails facing each other inside.	Thinking tails face water or phospholipids dissolve like salt.	Phospholipids placed in water.

Course scope (public description): Biochemistry covers the structure and function of the four major polymers produced by living organisms. These include nucleic acids, proteins, carbohydrates, and lipids. This course focuses on application and the underlying biochemistry in order to grasp how it is applied. This course will help students gain an introductory understanding of the chemicals and reactions that sustain life. Students will see the importance of this subject matter to health.